

Data Engineering Intern Technical Assessment

Amsterdam Airbnb Market Intelligence Report

Prepared for: Expernetic (Pvt) Ltd

Prepared by: Himaya Pathirage

Date: June 2026

Table of Contents

Table of Contents	2
1. Executive Summary.....	5
1.1 Key Findings	5
1.2 Top Recommendations	5
2. Objectives & Scope	5
2.1 Objective	5
2.2 City Selection.....	6
2.3 Scope and Prioritization	6
3. Dataset Overview	6
3.1 Source	6
3.2 Files Used	7
3.3 Entity Relationships.....	7
3.4 Business Domain Context	7
3.5 Key Dataset Limitations	8
3.6 Key Assumptions	8
4. Methodology.....	8
4.1 Overall Approach	8
4.2 Tooling.....	9
4.3 Key Methodological Decisions	9
4.4 Documentation Practice	9
5. Engineering Approach.....	10
5.1 Pipeline Overview	10
5.2 Data Quality Process	10
5.3 Data Model	10
5.4 Key Trade-offs	11
5.5 Pipeline Resilience	11
6. EDA Findings.....	11
6.1 Summary Statistics & Distributions.....	11
6.2 Geographic & Spatial Analysis.....	14
6.3 Temporal & Seasonal Trends	17
6.4 Host & Supply-Side Analysis.....	17
6.5 Review & Demand-Side Analysis.....	18
7. Statistical Findings.....	18

7.1 H1: Entire-Home vs. Private Room Pricing.....	18
7.2 H2: Superhost vs. Non-Superhost Review Scores.....	18
7.3 H3: Price by Review Count.....	19
7.4 H4: Neighbourhood Price Differences.....	19
7.5 H5: Weekend vs. Weekday Pricing.....	19
7.6 Correlation and Regression Analysis.....	19
7.7 Summary: Statistical vs. Practical Significance.....	20
8. Data Science Experiments.....	20
9. AI/ML Experiments.....	20
10. Visualizations.....	21
11. Business Recommendations.....	22
11.1 For Pricing Strategy.....	22
11.2 For Host Support and Segmentation.....	22
11.3 For Market Entry and Investment.....	22
11.4 For Demand Forecasting.....	22
11.5 A Caution on Calendar-Based Recommendations.....	23
12. Cross-City Comparisons.....	23
13. Limitations & Caveats.....	23
13.1 Dataset Limitations.....	23
13.2 Methodological Limitations.....	24
13.3 Scope Limitations.....	24
14. Future Improvements.....	24
14.1 With More Time.....	24
14.2 With Better Data.....	25
14.3 With Additional Resources.....	25
15. Reflection.....	25
15.1 How I Prioritized.....	25
15.2 Trade-offs I Made.....	25
15.3 Key Lessons Learned.....	26
15.4 What I Would Do Differently.....	26
Appendix A: AI Usage Disclosure.....	26
A.1 AI Tools Used.....	26
A.2 AI-Assisted Sections.....	26
A.3 Output Validation.....	27
A.4 Modifications Made.....	27

A.5 Critical Assessment	27
-------------------------------	----

1. Executive Summary

This report analyzes Amsterdam's short-term rental market using the Inside Airbnb dataset (10,480 listings, scraped September 2025). The analysis spans data engineering, exploratory analysis, and statistical testing to surface actionable findings for a market intelligence consultancy.

1.1 Key Findings

Property type is the single strongest price driver in this market. Entire-home listings command roughly 71% more than private rooms, a large, statistically confirmed effect (Mann-Whitney U, $p < 0.001$, rank-biserial correlation 0.622). Houseboats, a distinctly Amsterdam feature, command a further premium and cluster tightly along the city's canal network, a hard geographic constraint on this segment's growth.

The market is overwhelmingly composed of small, individual hosts: 93.7% of hosts manage exactly one listing, together accounting for 82.3% of total supply. Commercial, multi-listing operators remain a clear minority.

Review scores suffer from severe rating inflation, 75% of listings score 4.79 or higher out of 5, making the overall rating a poor differentiator. Breaking ratings into sub-dimensions reveals more useful signal: guests are most critical of value for money (mean 4.65) and cleanliness (4.78), while communication and check-in score consistently near the ceiling.

Demand has not just recovered from the COVID-19 disruption but grown beyond it: review volume (a proxy for bookings) fell sharply in 2020 before rebounding to exceed pre-pandemic levels by 2022, peaking in 2024.

Several data limitations meaningfully shape this analysis: calendar-level price data is entirely unusable for Amsterdam (100% null), and nearly 44% of listings have no published price, both are documented throughout this report rather than papered over.

1.2 Top Recommendations

Property type and capacity should anchor any pricing strategy; neighbourhood matters but is a secondary factor (explaining only ~7% of price variance); and review score should not be used as a quality signal given market-wide inflation, review count and sub-dimension scores are more informative.

2. Objectives & Scope

2.1 Objective

This project was undertaken as a technical assessment for the Data Engineer Intern role at Expernetic. The goal was to transform Inside Airbnb's raw, publicly available data into a structured analytics pipeline, statistically grounded findings, and actionable business recommendations for a hypothetical Airbnb market intelligence consultancy.

2.2 City Selection

Amsterdam was selected as the sole city for this analysis. This was the author's first industry-level data project, so the priority was to genuinely understand each step, ingestion, cleaning, modeling, analysis, and statistics, rather than to take on multiple cities just because it might look more impressive on paper. Going deep on one city felt more important than spreading effort thin across several and not understanding what was being done.

Amsterdam itself was chosen for practical reasons: it has a well-documented, moderately sized dataset (10,480 listings) with a long review history (spanning 2010-2025) and a distinctive feature, houseboat listings, that made the analysis more interesting and specific to this city rather than generic.

2.3 Scope and Prioritization

Given the assignment's explicit design philosophy, the following prioritization approach was taken:

- **Sections completed in full:** Dataset Familiarization (Section 2), Data Engineering (Section 3, including a PostgreSQL star schema), Exploratory Data Analysis (Section 4), and Statistical Analysis (Section 5).
- **Rationale:** These four sections form a complete, end-to-end analytical pipeline, from raw data to statistically validated business findings, and align directly with the core responsibilities of a Data Engineer Intern role: data ingestion, cleaning, modeling, and producing analysis-ready, trustworthy outputs.
- **Depth over breadth:** Within each completed section, effort was weighted toward investigating and resolving genuine ambiguities in the data (for example, the calendar price field's unusability, the superhost pricing paradox, and non-normal price distributions requiring non-parametric statistical tests) rather than superficially checking off every bullet point.
- **Optional sections:** As an intern, the priority was to do the mandatory and recommended sections (2 through 5) properly first, rather than spreading effort across every optional section. Data Science modeling (Section 6) and AI/LLM experimentation (Section 7) were not attempted for this reason; as someone early in this field, these felt like a bigger stretch to do well within the time available, compared to building further on the data engineering and analysis work already underway. If time remained after completing the core sections, Section 8 (Open Innovation) was preferred over Sections 6 and 7, since it offered more room to build something original within the author's current skill level.

3. Dataset Overview

3.1 Source

All data for this analysis comes from Inside Airbnb, a publicly available, independent dataset providing detailed information on Airbnb listings worldwide. The Amsterdam dataset used here was scraped on 2025-09-11.

3.2 Files Used

File	Rows	Columns	Purpose
listings.csv.gz	10,480	79	Core listing, host, and pricing data
calendar.csv.gz	3,825,200	7	Daily availability per listing
reviews.csv	501,084	2	Review dates per listing (summary)
reviews.csv.gz	~501,084	6	Detailed review text and reviewer IDs
neighbourhoods.csv	22	2	Neighbourhood names
neighbourhoods.geojson	22 features	-	Neighbourhood boundary polygons

3.3 Entity Relationships

listings.id serves as the primary key, joined to calendar.csv.gz and reviews files via listing_id, and to neighbourhoods.csv/.geojson via neighbourhood_cleansed. All relationships were verified directly: the 22 neighbourhood names match exactly across all three relevant files, and review counts computed independently from reviews.csv matched listings.csv.gz's own number_of_reviews column with zero discrepancies across all 10,480 listings.

3.4 Business Domain Context

This dataset represents the short-term rental marketplace in Amsterdam. Four entities make up the business model:

Listing: a single rental unit (entire apartment, private room, or shared room) published for guests to book, the basic unit of supply in this marketplace.

Host: the individual or business managing one or more listings. Hosts range from casual single-property owners to commercial operators managing dozens of units. Host attributes (response rate, superhost status, verification) function as trust signals for guests.

Review: guest feedback tied to a specific stay, serving as both a trust signal for future guests and the closest available proxy for actual booking activity, since the dataset contains no direct transaction data.

Neighbourhood: one of 22 geographic groupings used to segment the city for location-based analysis.

3.5 Key Dataset Limitations

- **Calendar price field is entirely unusable.** price and adjusted_price in calendar.csv.gz are 100% null across all 3.8 million rows, a scraping artifact specific to this city's batch. Revenue-related analysis relies instead on listings.csv.gz's pre-computed estimated_revenue_l365d.
- **Estimates, not measured outcomes.** estimated_occupancy_l365d and estimated_revenue_l365d are modeled using Inside Airbnb's own methodology, which assumes a fixed review-to-booking conversion rate. These are directional estimates, not confirmed transactions.
- **Single snapshot, not a continuous feed.** All listing-level data reflects one scrape date; price history and past availability are not captured.
- **No guest-side or financial data.** No demographics, booking dates, or real revenue/cost figures exist anywhere in the dataset.
- **Significant missingness in core fields.** Price is missing for 44.0% of listings; beds and bathrooms are each missing for roughly 43-44%. These gaps are treated as explicit nulls throughout this analysis, never imputed, to avoid fabricating values that don't exist.

3.6 Key Assumptions

- Null price values are treated as “no price published,” not as a free listing, and are excluded from price-based statistics rather than imputed.
- availability_365 is never used alone as a demand signal, since low availability can reflect either genuine bookings or a host manually blocking dates, a distinction this dataset cannot resolve.
- neighbourhood_cleansed (Inside Airbnb's standardized field) is used exclusively over the raw, 50.5%-null neighbourhood field.
- Review counts are treated as a lower-bound proxy for bookings, not an exact count, since not every guest leaves a review.

A complete, detailed account of every dataset finding, assumption, and engineering decision is maintained in the project's decisions log (docs/decisions_log.md), included with the submitted source code.

4. Methodology

4.1 Overall Approach

This analysis followed a structured, four-stage pipeline: dataset familiarization, data engineering, exploratory data analysis, and statistical testing. Each stage was treated as a foundation for the next, schema documentation (Section 2) informed cleaning decisions (Section 3.2), which in turn shaped what enrichment and modeling were possible (Sections 3.3-3.4), which then fed directly into exploratory analysis (Section 4) and formal hypothesis testing (Section 5).

4.2 Tooling

Python (pandas, NumPy) was used for data ingestion, cleaning, and enrichment. PostgreSQL 18 was used to implement a star schema for analytical querying, chosen over alternatives like DuckDB or SQLite due to prior hands-on experience, reducing setup risk given the project's one-week timeline. Statistical testing used scipy and statsmodels. Visualization used matplotlib, seaborn, and folium (for geospatial mapping).

4.3 Key Methodological Decisions

Verification over assumption. Throughout this project, claims were checked against the actual data before being treated as fact. For example, an initial assumption that calendar data could support a revenue model was abandoned only after directly confirming the relevant fields were 100% null, and a counterintuitive finding (non-superhosts pricing higher than superhosts) was investigated further rather than reported at face value, revealing a property-type confound that explained the pattern.

Non-parametric statistics where appropriate. Price and review-related variables showed strong skew and, in some cases, unequal variance across groups (Section 4.1, Section 7). While large sample sizes can sometimes make standard t-tests and ANOVA reasonably robust even without normally distributed data, through the Central Limit Theorem, the severity of skew in this dataset (for example, entire-home prices having a standard deviation larger than their own mean) and confirmed unequal variances across neighbourhoods (Levene's test) made non-parametric alternatives, Mann-Whitney U and Kruskal-Wallis, the safer, more defensible choice throughout Section 5.

Outliers flagged, never silently deleted. Confirmed data errors (for example, listings priced above EUR 40,000 per night, traced to specific data entry mistakes and test listings) were marked with a boolean flag and excluded from relevant statistics, while the raw data itself was always preserved unchanged.

Effect size alongside significance. Every statistical test in Section 5 reports an effect size (rank-biserial correlation, eta-squared, or Cohen's d) alongside its p-value, since statistical significance alone does not indicate whether a finding is practically important, a distinction made explicit throughout this report.

4.4 Documentation Practice

Every meaningful decision, including tool choices, cleaning rules, and responses to unexpected findings, was logged in real time in a running decisions log (docs/decisions_log.md), recording the options considered, the reasoning for the choice made, and the trade-off accepted. This log is included as a primary submitted artifact and should be read alongside this report for full traceability of the analytical process.

5. Engineering Approach

5.1 Pipeline Overview

A modular Python pipeline was built across four scripts, each with a single responsibility:

- **ingest.py**: loads raw files (listings, calendar, reviews, neighbourhoods) from disk with basic error handling.
- **clean.py**: standardizes price (removing currency symbols, casting to numeric), parses date fields, normalizes property types into 7 consistent categories, flags missing core attributes explicitly, applies domain validation rules, and standardizes coordinate precision.
- **enrich.py**: joins listings with review counts and calendar-derived occupancy rates, computes neighbourhood-level aggregates, and derives host tenure, review frequency, and price-per-bedroom.
- **load_to_postgres.py**: builds and loads a star schema into PostgreSQL.

Each script can be run independently and reuses functions from earlier stages, avoiding duplicated logic across the pipeline.

5.2 Data Quality Process

A formal data quality report was generated programmatically (docs/data_quality_report.md), profiling row counts, duplicate records, and missingness across every ingested file. This surfaced several non-obvious findings that shaped downstream decisions:

- Calendar price fields, expected to support revenue analysis, were found to be 100% null, redirecting that analysis to listings.csv.gz's pre-computed estimates instead.
- An automated duplicate check flagged 27,545 rows in reviews.csv; manual cross-referencing against the detailed reviews file confirmed these were genuine, distinct reviews rather than data errors, a structural artifact of that file's limited columns, not a quality problem.
- Seven listings were confirmed, through manual inspection of listing names, to carry data-entry-error prices (up to EUR 80,018 per night) and were flagged, not deleted, preserving the raw dataset while excluding errors from statistical analysis.

5.3 Data Model

A star schema was implemented in PostgreSQL 18, consisting of one fact table and three dimension tables:

- **fact_listings** (grain: one row per listing): pricing, occupancy, review, and derived metrics, with foreign keys to each dimension.
- **dim_host**: one row per unique host (9,201 rows).

- **dim_neighbourhood**: one row per neighbourhood (22 rows), with pre-computed median price, listing count, and average rating.
- **dim_property_type**: one row per grouped property type (7 rows).

Four analytical SQL queries were written and validated against this model (src/queries.sql), covering top neighbourhoods by price, superhost pricing comparison, price by property type, and occupancy by review-count bucket, all confirmed to return results consistent with independent Python-side calculations.

5.4 Key Trade-offs

- **Denormalization in dim_neighbourhood**: aggregates are pre-computed for fast querying, at the cost of becoming a snapshot that would need to be rebuilt if underlying listing data changed.
- **No slowly changing dimension handling**: dim_host reflects only the current state, appropriate given Inside Airbnb itself provides only a single snapshot, not a historical feed.
- **Listing-grain fact table only**: a second, calendar-grain fact table (3.8 million rows) was not built, given the project timeline and the limited additional analytical value once calendar's price field was confirmed unusable.
- **PostgreSQL over DuckDB or SQLite**: chosen for prior hands-on familiarity, accepting the added setup overhead of a full database server.

5.5 Pipeline Resilience

Basic error handling was added to ingestion functions, and the pipeline's data lineage (which transformations produce each output table, from which source files) is documented in docs/data_model.md. Given the single-snapshot nature of this dataset, a full incremental-processing system was not built; the approach that would be used for new scraped data is described in that same document rather than implemented, a deliberate scope decision given the one-week timeline.

6. EDA Findings

6.1 Summary Statistics & Distributions

The typical Amsterdam listing accommodates 2 guests with 1 bedroom, priced at a median of EUR 222 per night (the mean, EUR 336.79, is pulled upward by a small number of extreme high prices and should not be used for typical-market benchmarking).

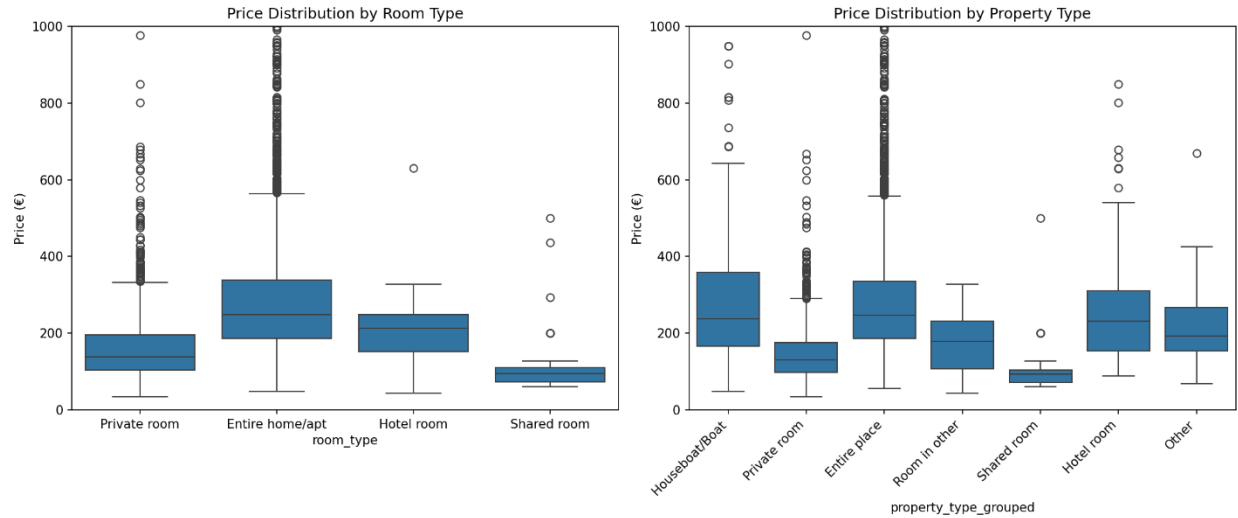


Figure 1: Price Distribution by Room Type and Property Type

Source: reports/price_by_type.png

Entire-home listings command roughly double the price of private rooms and triple that of shared rooms. Houseboats command a modest premium even over standard entire-place listings, reflecting their status as a distinctive, Amsterdam-specific draw.

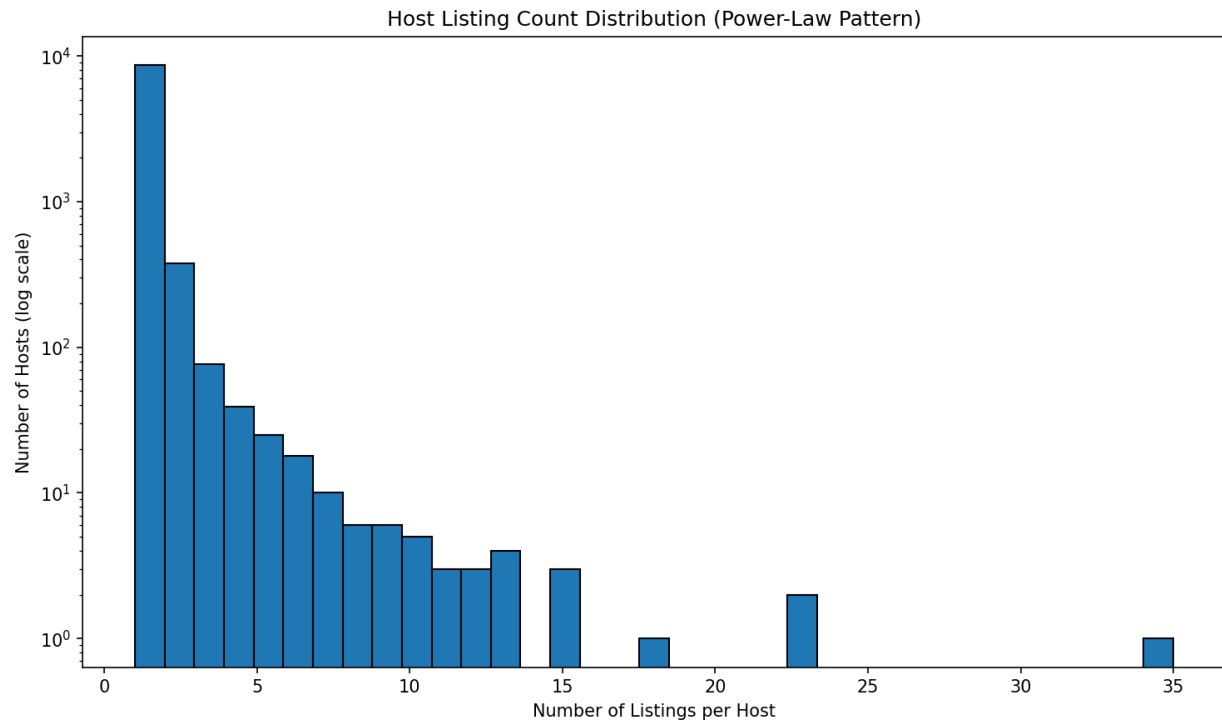


Figure 2: Host Listing Count Distribution

Source: reports/host_listing_power_law.png

Amsterdam's market follows a clear power-law pattern: 93.7% of hosts manage exactly one listing, together holding 82.3% of total supply. A small group of commercial-style operators exists (the largest single host manages 35 listings), but this group remains a minority of overall supply.

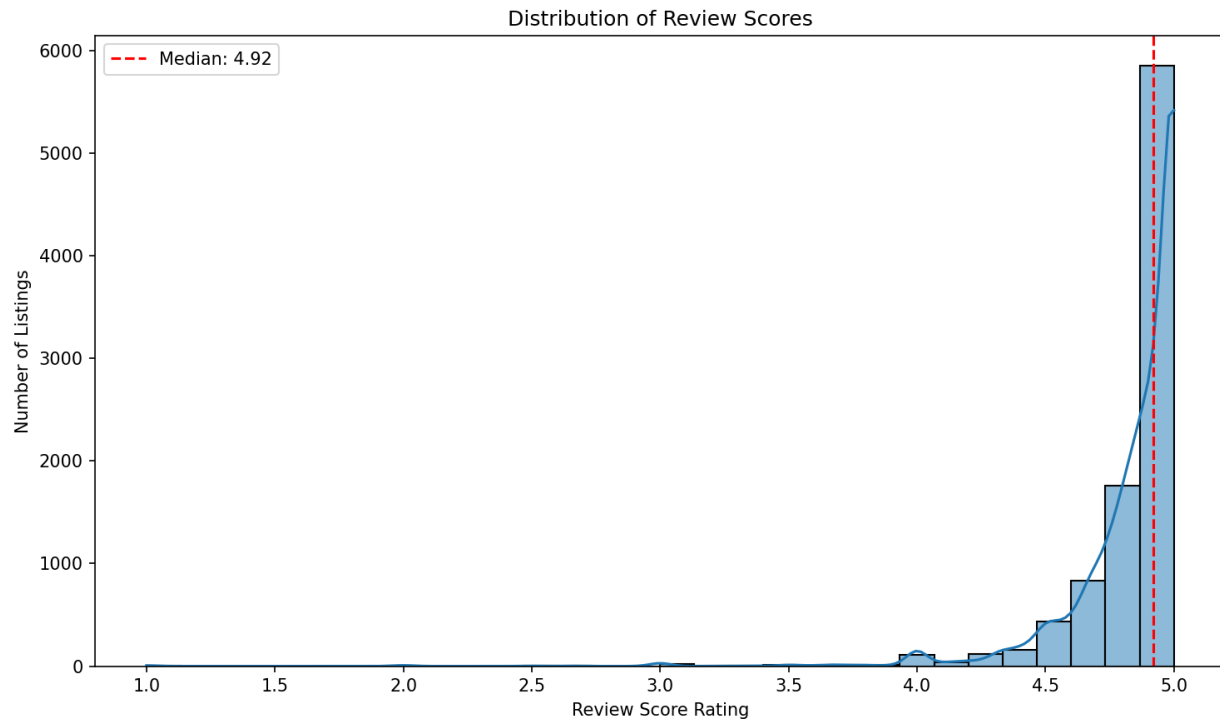


Figure 3: Review Score Distribution

Source: [reports/review_score_distribution.png](#)

Review scores show severe rating inflation: the median score is 4.92 out of 5, and 75% of listings score 4.79 or higher. Scores below 4.0 are vanishingly rare, meaning the overall rating provides very little practical ability to distinguish listing quality.

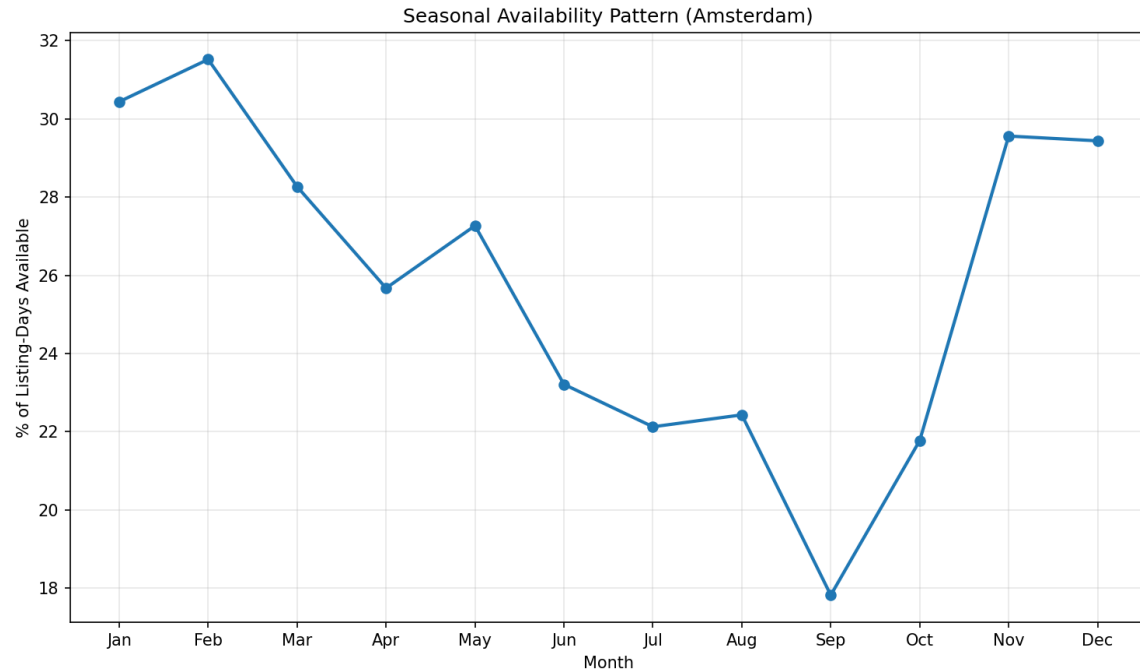


Figure 4: Seasonal Availability Pattern

Source: reports/seasonal_availability.png

Availability is lowest during the summer tourist season (around 22% in June through August) and highest in winter (around 30-32%), consistent with known Amsterdam tourism patterns, though this metric cannot distinguish genuine bookings from host-blocked dates.

6.2 Geographic & Spatial Analysis

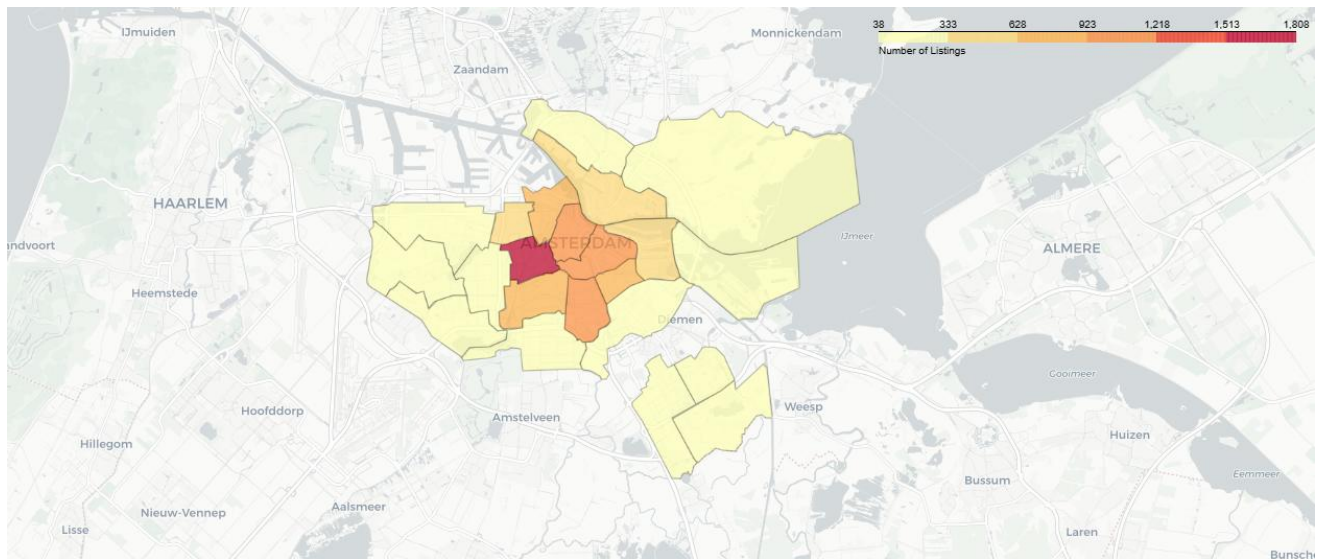


Figure 5: Listing Density Map

Source: reports/listing_density_map.html (static capture)

Listing supply is heavily concentrated in central Amsterdam, fading progressively toward the city's edges.

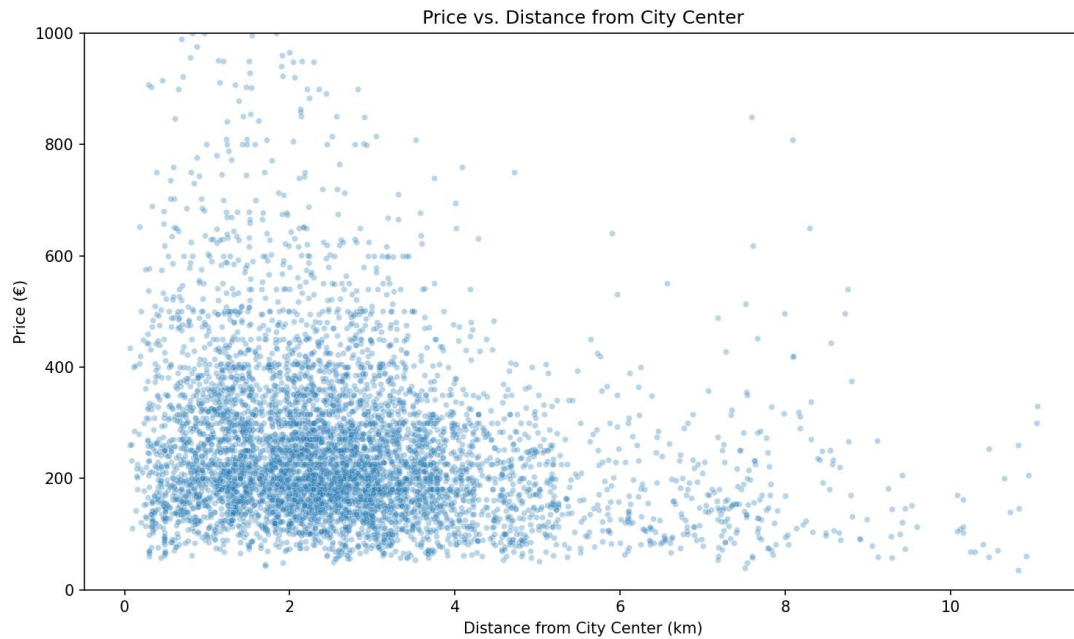


Figure 6: Price vs. Distance from City Center

Source: reports/price_vs_distance.png

Distance from the city center shows essentially no linear correlation with price ($r = 0.022$). The highest-priced listings cluster within the first 1-2km of the center, but plenty of low-priced listings exist in that same zone, meaning price variation near the center is driven by property type and quality, not fine-grained location.

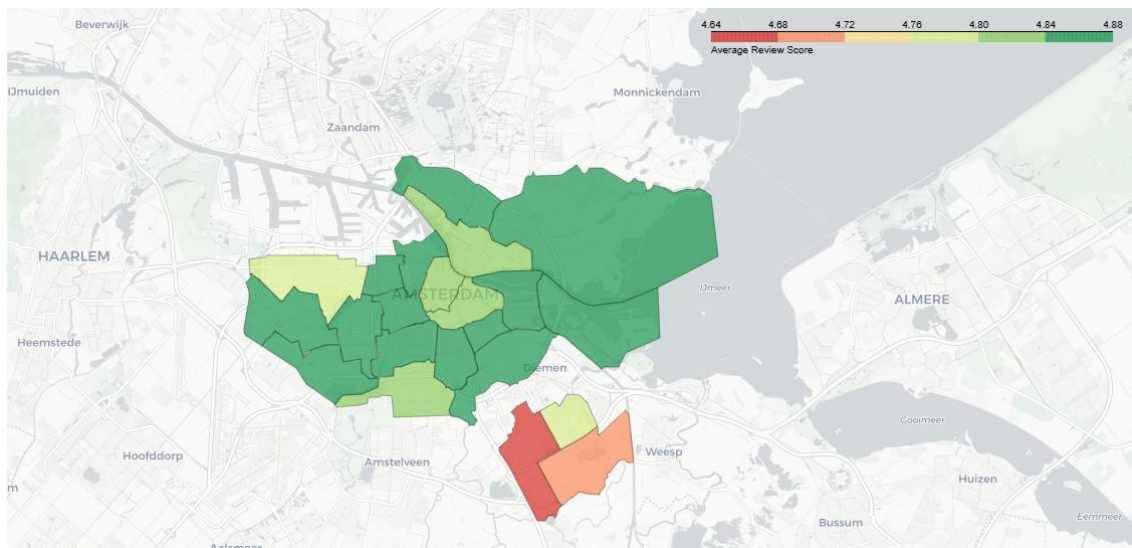


Figure 7: Average Review Score Map

Source: reports/review_score_map.html (static capture)

While most neighbourhoods cluster tightly between 4.80 and 4.88, two neighbourhoods in Amsterdam's southeastern Bijlmer district stand out as genuinely lower-scoring (Bijlmer-Centrum, 4.64; Gaasperdam - Driemond, 4.71), based on sample sizes large enough to rule out random noise.

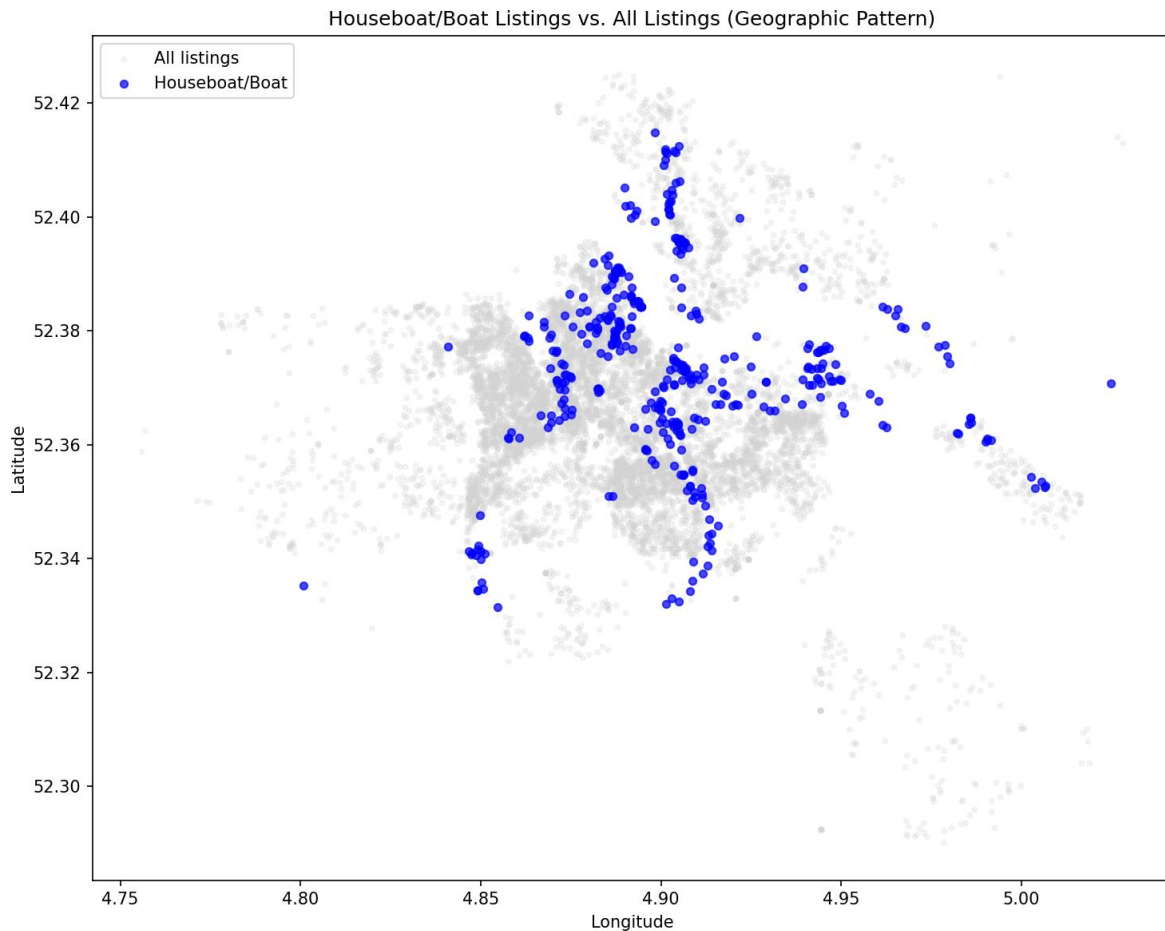


Figure 8: Houseboat/Boat Geographic Clustering

Source: reports/houseboat_clustering.png

Houseboat listings trace clear linear and curved patterns across the map, almost certainly corresponding to Amsterdam's canal network, confirming this property type is geographically constrained rather than freely expandable.

6.3 Temporal & Seasonal Trends

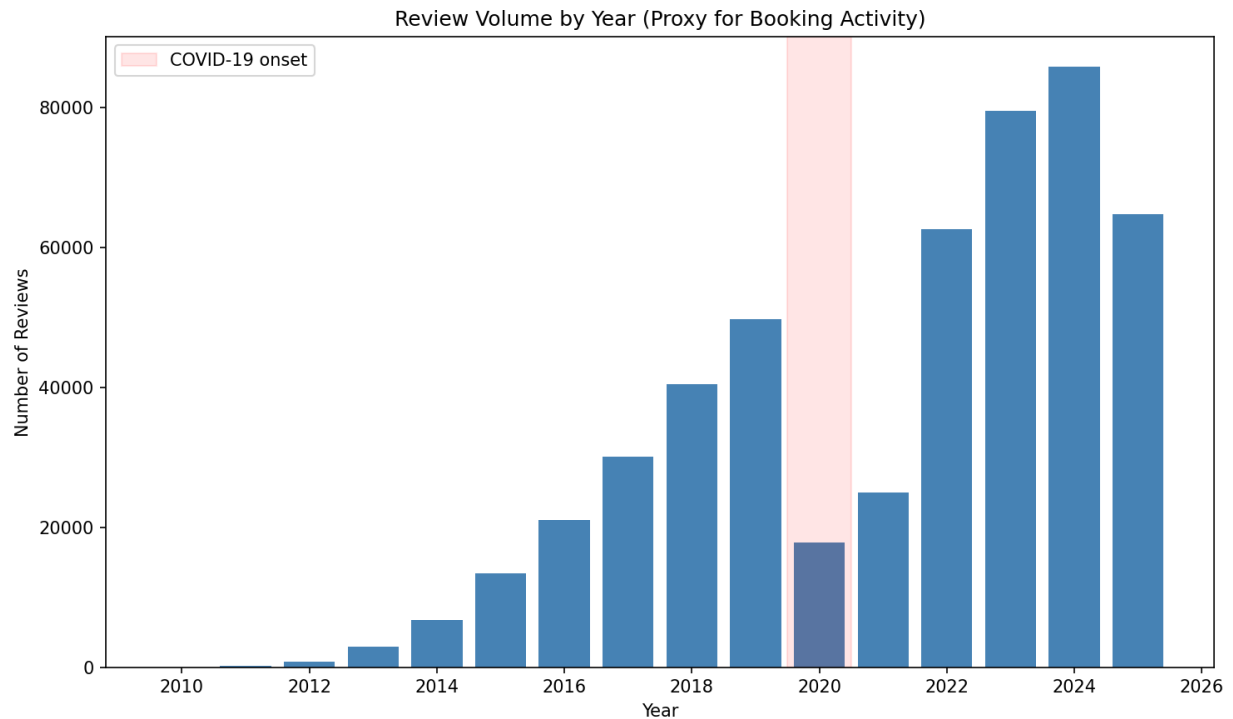


Figure 9: Review Volume by Year

Source: *reports/review_volume_by_year.png*

Review volume grew steadily from 2010 through 2019, collapsed sharply in 2020 (COVID-19), and not only recovered but substantially exceeded pre-pandemic levels by 2022-2024, peaking at 85,885 reviews in 2024.

Host tenure showed only a weak relationship with pricing; newer hosts (0-2 years) priced slightly lower (median EUR 213) than more established hosts (EUR 222-232), but the gap was modest. A distinct long-term-stay market segment was identified: 71 listings (0.7%) require minimum stays over 30 nights, confirmed via listing names to be genuine extended-stay properties rather than data errors. True seasonal pricing analysis could not be performed, as calendar-level price data is unusable for this dataset (Section 3.5).

6.4 Host & Supply-Side Analysis

A counterintuitive finding, non-superhosts pricing higher on average (EUR 277.07) than superhosts (EUR 256.15), was investigated and explained: superhosts run a markedly different property type mix (36.7% private rooms vs. 7.8% for non-superhosts), and since private rooms price lower across the market overall, this composition difference, not superhost status itself, accounts for the gap.

A similar pattern appeared between commercial (multi-listing) and casual (single-listing) hosts, with casual hosts pricing higher on average, partly explained by commercial hosts running a more diversified portfolio including lower-priced hotel-room and private-room listings.

6.5 Review & Demand-Side Analysis

Review count, review score, and price show only weak pairwise correlations (all between -0.09 and -0.05), consistent with the rating inflation pattern limiting review score's usefulness as a signal. Listings combining high review counts with below-average scores were found to skew heavily toward private rooms and hotel rooms (80% of this group), at a median price well below the overall market.

Breaking review scores into sub-dimensions revealed the report's most actionable finding: value-for-money scores lowest (mean 4.65) and showed the widest spread of any sub-dimension, while communication (4.91) and check-in (4.89) scored highest and most consistently. Guests are most critical of whether a stay felt worth its price, not of host responsiveness.

7. Statistical Findings

All hypothesis tests in this section used non-parametric methods (Mann-Whitney U, Kruskal-Wallis) rather than standard t-tests or ANOVA. While large sample sizes can sometimes make standard tests reasonably robust even without normally distributed data, through the Central Limit Theorem, the severity of skew in price data and the confirmed unequal variances across neighbourhood groups (Levene's test, Section 7.4) made non-parametric tests the more defensible choice here, full reasoning for each test is given alongside its results below.

7.1 H1: Entire-Home vs. Private Room Pricing

Entire-home listings (mean EUR 301.00, $n=4,530$) were tested against private-room listings (mean EUR 175.73, $n=1,274$) using a Mann-Whitney U test. The result was highly significant ($p = 4.95 \times 10^{-253}$) with a large effect size (rank-biserial correlation = 0.622). The null hypothesis is rejected: entire-home listings are priced significantly higher than private rooms, by a margin that is both statistically robust and practically substantial, roughly 71% higher on average.

7.2 H2: Superhost vs. Non-Superhost Review Scores

Superhost listings (mean 4.865, $n=1,813$) were tested against non-superhost listings (mean 4.839, $n=7,460$). The result was significant ($p = 2.01 \times 10^{-18}$) but the effect size was small (rank-biserial correlation = 0.128). The null hypothesis is rejected, but the actual score difference (0.026 points on a 5-point scale) is too small to be practically meaningful. This is a clear example of statistical significance not implying practical importance, a large sample size makes even tiny differences detectable.

7.3 H3: Price by Review Count

Listings with more than 10 reviews (mean EUR 257.32, $n=3,054$) were tested against listings with 10 or fewer reviews (mean EUR 288.79, $n=2,813$). The result was significant ($p = 6.98 \times 10^{-25}$) with a small effect size (0.155). Counterintuitively, listings with more reviews are priced lower on average, likely because cheaper listings book more frequently and accumulate reviews faster, not because more reviews indicate a premium listing.

7.4 H4: Neighbourhood Price Differences

Price was compared across all 22 neighbourhoods using a Kruskal-Wallis test (the non-parametric equivalent of one-way ANOVA, selected after Levene's test confirmed unequal variances across groups). The result was significant ($p = 2.51 \times 10^{-75}$) with a small-to-medium effect size (eta-squared = 0.068). Neighbourhood has a real, confirmed effect on price, but explains only about 7% of total price variation, a far smaller driver than property type (Section 7.1).

Confidence interval analysis further showed that while overall neighbourhood differences are significant, not every pair of neighbourhoods is distinguishable: Centrum-West, Centrum-Oost, and Zuid have overlapping 95% confidence intervals, while Centrum-West and De Baarsjes - Oud-West do not, indicating genuine pairwise differences exist only between some neighbourhoods, not all.

7.5 H5: Weekend vs. Weekday Pricing

This hypothesis could not be tested. It requires day-level price data from calendar.csv.gz, which was confirmed 100% null for Amsterdam across all 3.8 million calendar rows (Section 3.5). This is a genuine dataset limitation, not an analytical shortcut, and is documented here transparently rather than answered with fabricated results.

7.6 Correlation and Regression Analysis

Capacity-related variables (bedrooms, accommodates, beds) were the strongest numerical correlates with price ($r = 0.23$ to 0.27); review score, host tenure, and minimum nights showed negligible correlation. A multiple regression using these and other numerical features explained only 9.6% of price variation ($R\text{-squared} = 0.096$), confirming that most pricing variation comes from factors outside this set, chiefly property type, as shown in Section 7.1.

A variance inflation factor (VIF) check found no severe multicollinearity among capacity variables (all VIF values below 3.2), though beds showed an unexpected negative coefficient once bedrooms and accommodates were included, a likely sign of moderate overlap between these variables rather than a genuine pricing effect.

A LOWESS smoothing curve revealed that the relationship between accommodates and price is non-linear: price rises steeply from 1 to 4 guests, flattens between 6 and 10 guests, then rises again at the high end (12+ guests), indicating a standard linear regression coefficient understates and overstates the true relationship at different points in the range.

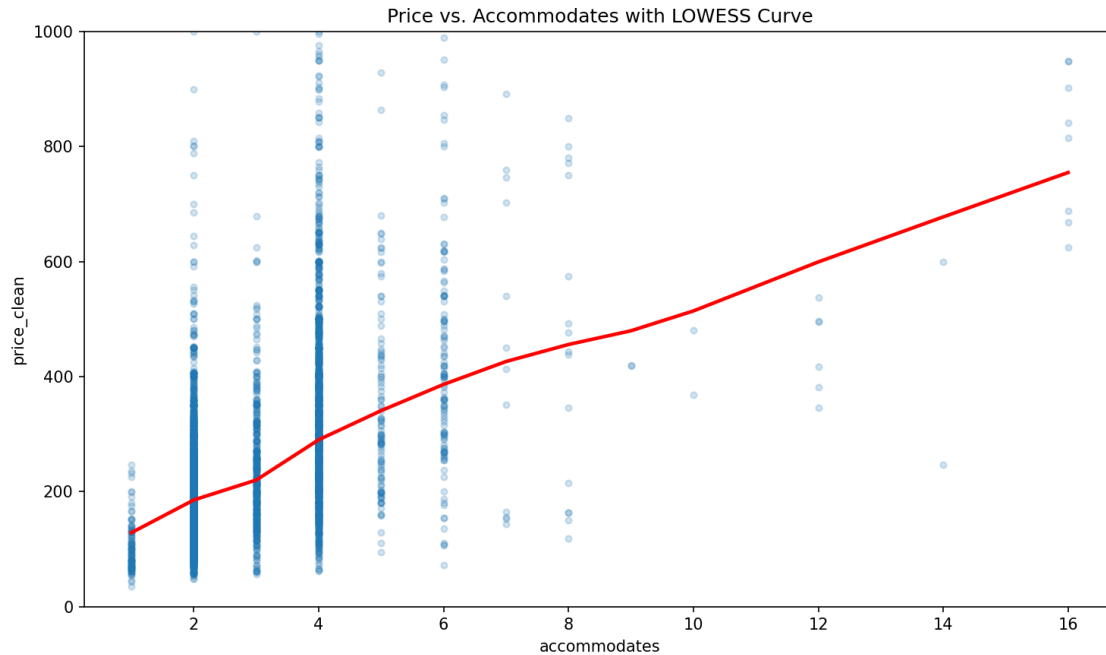


Figure 10. Price vs. Accommodates with LOWESS Curve.

Source: reports/price_vs_accomodates_lowess.png

7.7 Summary: Statistical vs. Practical Significance

Every tested hypothesis was statistically significant, but effect sizes varied widely, from 0.622 (large, H1) down to 0.068 (small-medium, H4). With a sample of 10,480 listings, even small, practically minor differences become statistically detectable. Property type (H1) is the only finding in this analysis large enough to be a primary pricing driver on its own; the remaining findings are real but secondary factors.

8. Data Science Experiments

Not attempted in this submission. As explained in Section 2.3 (Objectives & Scope) and Section 13.3 (Scope Limitations), priority was given to thoroughly completing the mandatory and recommended sections of the assignment (dataset familiarization, data engineering, EDA, and statistical analysis) before considering optional, more advanced work. Section 14.1 (Future Improvements) outlines what modeling work would be pursued with more time, particularly price prediction informed by the non-linear capacity relationship found in Section 7.6.

9. AI/ML Experiments

Not attempted in this submission, for the same reasons described in Section 8 above. AI tools were used throughout this project as a development aid (see Appendix A: AI Usage Disclosure), but no standalone NLP, LLM, or agentic experimentation was conducted as part of the analytical deliverables themselves.

10. Visualizations

This section provides a consolidated index of all figures referenced throughout this report, with captions and source notes. Full-size versions appear inline at their first reference in Section 6 (EDA Findings).

Figure 1. Price Distribution by Room Type and Property Type. Boxplots showing price spread across room types and grouped property types, outliers (confirmed data errors) excluded. Source: reports/price_by_type.png.

Figure 2. Host Listing Count Distribution (Power-Law Pattern). Log-scale histogram of listings per host, illustrating that 93.7% of hosts manage exactly one listing. Source: reports/host_listing_power_law.png.

Figure 3. Distribution of Review Scores. Histogram with density curve showing severe rating inflation, the large majority of listings score between 4.5 and 5.0. Source: reports/review_score_distribution.png.

Figure 4. Seasonal Availability Pattern (Amsterdam). Line chart of monthly availability percentage, showing lower availability during summer tourist months. Source: reports/seasonal_availability.png.

Figure 5. Listing Density Map. Interactive choropleth map of listing counts by neighbourhood, darkest shading indicating highest density in central Amsterdam. Source: reports/listing_density_map.html (static capture included in report).

Figure 6. Price vs. Distance from City Center. Scatter plot showing no clear linear relationship between proximity to the center and price. Source: reports/price_vs_distance.png.

Figure 7. Average Review Score Map. Choropleth map by neighbourhood, showing two notably lower-scoring areas in the southeastern Bijlmer district. Source: reports/review_score_map.html (static capture included in report).

Figure 8. Houseboat/Boat Listings vs. All Listings. Scatter plot isolating houseboat locations, revealing canal-network clustering patterns. Source: reports/houseboat_clustering.png.

Figure 9. Review Volume by Year. Bar chart of yearly review counts (2010-2025), highlighting the COVID-19 disruption and subsequent recovery. Source: reports/review_volume_by_year.png.

Figure 10. Price vs. Accommodates with LOWESS Curve. Scatter plot with a non-parametric smoothing curve revealing a non-linear price-capacity relationship. Source: reports/price_vs_accommodates_lowess.png.

All figures were generated using matplotlib, seaborn, or folium, and saved at 150 DPI for print quality. Source code for each figure is available in notebooks/02_edu.ipynb and notebooks/03_statistics.ipynb.

11. Business Recommendations

11.1 For Pricing Strategy

Anchor pricing decisions on property type and capacity, not location alone. Property type explains far more price variation (rank-biserial effect size 0.622) than neighbourhood does (eta-squared 0.068). A host or platform advising on pricing should prioritize guidance around property type and guest capacity first, and treat neighbourhood as a secondary adjustment factor.

Recognize the non-linear value of added capacity. The LOWESS analysis (Section 7.6) shows that capacity gains in the 6-10 guest range add comparatively little price premium, while the jump from 1 to 4 guests and the jump above 12 guests both carry disproportionately higher returns. Hosts considering property expansion should weigh this curve rather than assuming a constant per-guest price increase.

11.2 For Host Support and Segmentation

Do not use review score as a host quality filter. Severe rating inflation (75% of listings score 4.79 or higher) means the overall review score cannot meaningfully distinguish host performance. Any host-quality segmentation should instead use review sub-dimension scores, particularly value-for-money and cleanliness, which show genuine variation and likely reflect real differences in guest experience.

Treat superhost status as a behavioral signal, not a pricing or quality signal. Superhost status correlates with a different property type mix (more private rooms) rather than meaningfully higher prices or scores. It is better understood as a proxy for hands-on, individually managed hosting style, useful for understanding host segments, not for predicting listing value.

11.3 For Market Entry and Investment

Central neighbourhoods offer both the highest demand and the highest competition. Listing density and price both peak in the same central neighbourhoods (Centrum-West, De Pijp - Rivierenbuurt), indicating strong, sustained demand that supports premium pricing despite high supply. New entrants should expect to compete directly on quality and differentiation in these areas, rather than relying on under-supply.

Houseboat listings represent a structurally limited but premium niche. This segment commands the highest average price of any property type but is constrained by the physical extent of Amsterdam's canal network. Growth strategies should not assume this segment can scale the way standard apartment listings can.

11.4 For Demand Forecasting

The market has structurally grown beyond pre-pandemic levels. Review volume, used as a demand proxy, did not just recover from COVID-19 but substantially exceeded 2019 levels by 2022-2024. Forecasts and capacity planning should treat 2019 as an outdated baseline, not a ceiling.

Investigate the southeastern Bijlmer area's lower guest satisfaction scores. With sample sizes large enough to rule out noise, this area shows a genuine, persistent rating gap. This may represent either an opportunity (under-served improvement potential) or a risk (structurally lower guest fit for short-term rental in that area), and warrants qualitative follow-up before drawing further conclusions.

11.5 A Caution on Calendar-Based Recommendations

Any recommendation requiring day-level pricing or weekday/weekend booking patterns cannot currently be supported by this dataset, since Amsterdam's calendar price data is entirely unusable (Section 3.5). Stakeholders should be aware that demand timing within the week or specific calendar dates remains an open question this analysis cannot answer.

12. Cross-City Comparisons

Not applicable. This analysis covers Amsterdam only, a deliberate scope decision explained in Section 2.2 (City Selection). No comparative, multi-market data was collected or analyzed.

13. Limitations & Caveats

13.1 Dataset Limitations

Calendar price data is unusable. The price and adjusted_price fields in calendar.csv.gz are 100% null across all 3.8 million rows for Amsterdam. This is a known limitation of this specific scrape batch, not an analytical choice, and it directly prevented testing H5 (weekend vs. weekday pricing, Section 7.5) and any true seasonal pricing analysis (Section 6.3).

Significant missingness in core fields. Price is missing for 44.0% of listings; beds and bathrooms are each missing for roughly 43-44%. All price-based statistics in this report are computed on the subset of listings with published prices (approximately 56% of the dataset), which may not be representative of the full market, inactive or paused listings may disproportionately lack a public price.

Estimates, not measured outcomes. estimated_occupancy_1365d and estimated_revenue_1365d rely on Inside Airbnb's own modeling methodology, which assumes a fixed review-to-booking conversion rate. This conversion rate is itself debated in independent research. These figures should be read as directional estimates, not verified transactions.

Single point-in-time snapshot. All listing-level data reflects one scrape date (2025-09-11). No historical price or availability trend exists within this dataset for any individual listing; only review dates provide a genuine multi-year time series (2010-2025).

No guest-side or financial data. The dataset contains no guest demographics, exact booking transactions, or real revenue/cost figures. Demand-side conclusions throughout this report rely on review activity as an imperfect proxy.

13.2 Methodological Limitations

Occupancy is a proxy, not a confirmed booking measure. `occupancy_rate_calculated`, derived from the calendar's available field, cannot distinguish a listing that is genuinely booked from one where the host has manually blocked the calendar. This ambiguity affects every finding in this report that references occupancy or availability.

Correlational, not causal, findings. All relationships reported in Sections 6 and 7 (for example, property type and price, or neighbourhood and price) are statistical associations observed in a single snapshot of data. None of the analysis in this report supports causal claims, only Section 7's regression attempted to control for multiple factors simultaneously, and even that explained just 9.6% of price variation, leaving most of the underlying causal structure unexplained.

Regression assumptions only partially met. While the OLS regression in Section 7.6 passed a VIF multicollinearity check, the residual diagnostics (Jarque-Bera test, reported in the model summary) indicate the residuals are not normally distributed, consistent with price's known skew. Coefficient estimates should be treated as approximate rather than precise.

Single-city scope. All findings in this report are specific to Amsterdam and should not be assumed to generalize to other cities' short-term rental markets without separate validation.

13.3 Scope Limitations

Optional sections of the original assignment brief, Data Science modeling (Section 6), AI/LLM experimentation (Section 7), and (subject to final time available) the Open Innovation challenge (Section 8), were deprioritized in favor of thoroughly completing the mandatory and recommended sections. This was a deliberate choice (Section 2.3), not an oversight, but it means this report does not include predictive modeling, NLP analysis of review text, or experimental AI-driven features.

14. Future Improvements

14.1 With More Time

Attempt the deprioritized optional sections. Given more time, Data Science modeling (Section 6 of the brief, particularly price prediction with proper handling of the non-linear capacity relationship found in Section 7.6) and NLP analysis of review text (Section 7 of the brief) would add genuine value, especially since review text was identified throughout this report as a richer signal than the numeric review score.

Resolve the beds coefficient anomaly. The unexpected negative coefficient for beds in the regression model (Section 7.6) was flagged as tentative. With more time, this would be investigated further using interaction terms or a model less sensitive to overlap between capacity-related variables.

Build the calendar-grain fact table. A second fact table at the listing-per-day grain (deferred in Section 5.4 as a scope trade-off) would enable proper seasonal and weekday/weekend analysis using the available field, even without usable price data.

14.2 With Better Data

Request or source day-level price history. If a calendar snapshot with populated price data were available (from a different scrape batch or city), H5 (weekend vs. weekday pricing) and true seasonal pricing trends could finally be tested directly.

Obtain actual booking or transaction data. Every demand-side finding in this report relies on review counts as a proxy. Real booking data would remove the review-rate assumption entirely and allow direct occupancy and revenue measurement rather than estimation.

Multiple scrape snapshots over time. A single point-in-time snapshot cannot show how an individual listing's price or availability changes month to month. Repeated scrapes over a year would enable genuine within-listing seasonal analysis, rather than the cross-sectional comparisons used throughout this report.

14.3 With Additional Resources

Investigate the Bijlmer rating gap qualitatively. The lower review scores found in Amsterdam's southeastern Bijlmer district (Section 6.2) warrant follow-up beyond what this dataset can support, for example, guest interviews or review text analysis specific to that area, to understand whether this reflects a genuine service gap or a mismatch between guest expectations and that area's housing stock.

Validate findings against a second city. Testing whether the patterns found here (rating inflation, the superhost pricing paradox, the weak distance-price relationship) hold in a different market would clarify which findings are Amsterdam-specific versus broader Airbnb platform dynamics.

15. Reflection

15.1 How I Prioritized

This being my first industry-level data project, my main goal was to genuinely understand every step rather than maximize apparent scope. I chose a single city over multiple, and within that scope, I focused entirely on the mandatory and recommended sections (dataset familiarization, data engineering, EDA, and statistical analysis) before considering anything optional. Within each section, I tried to follow findings wherever they led, even when that meant slowing down to investigate something unexpected (like the calendar price field being unusable, or the superhost pricing pattern going against intuition) rather than rushing past it to cover more ground.

15.2 Trade-offs I Made

I accepted a smaller analytical footprint in exchange for depth and correctness. For example, I chose not to build a calendar-grain fact table even though it would have enabled richer seasonal analysis, because the core price field it would have relied on was already confirmed unusable, building it anyway would have added engineering complexity without a clear analytical payoff given my timeline. Similarly, I chose

PostgreSQL over a simpler tool like DuckDB because I already had some experience with it, prioritizing reducing setup risk over experimenting with a new tool this week.

I also made a deliberate choice throughout to flag data issues rather than silently fix or hide them, even when that meant admitting a hypothesis (H5) simply could not be tested with the data available.

15.3 Key Lessons Learned

Verifying output matters as much as writing code. More than once this week, code that ran without errors still produced incomplete or incorrect results, for example, a missing-value flag column that I had documented as a decision but forgot to actually include when building the database table. Catching this required checking the real output in pgAdmin, not just trusting that the script printed success.

Statistical significance and practical significance are genuinely different questions. Several of my hypothesis tests came back statistically significant with very small effect sizes. Learning to report and explain that distinction, rather than treating every significant p-value as an equally important finding, was one of the most useful parts of this project.

A surprising result is a prompt to investigate, not a finding to report immediately. The superhost pricing pattern initially looked like a strange anomaly, but tracing it back to a property-type confound turned it into one of the more interesting findings in this report. I learned to treat unexpected results as a starting point for further checking rather than either dismissing them or reporting them at face value.

15.4 What I Would Do Differently

Looking back, I would build a shorter, rougher version of the final report earlier in the week, rather than only beginning to assemble it after finishing all the analysis. Several sections, in particular the decisions log, grew large enough that consolidating everything into report form at the end took longer than I expected.

Appendix A: AI Usage Disclosure

A.1 AI Tools Used

Claude (Anthropic), accessed via the Claude.ai chat interface, was the only AI tool used throughout this project.

A.2 AI-Assisted Sections

This was my first industry-level data project, and I went into it with real gaps in practical knowledge. I had studied statistical testing methods (Mann-Whitney U, Kruskal-Wallis, ANOVA, and related concepts) and basic database concepts during my undergraduate studies, but I had not yet applied most of this in a real, end-to-end project before. Most of the practical side, how to actually structure a data engineering pipeline, how to investigate a messy real-world dataset, how to decide between competing approaches, was new to me going in.

I used Claude across every section of this project to understand what each part of the assignment was actually asking for, what concepts and tools were relevant, and what steps I should take next. But I did not treat its suggestions as instructions to follow blindly. I questioned its reasoning throughout, asked it to explain things I did not understand, pushed back when something did not make sense to me or did not match how I wanted to express a finding, and verified outputs myself before accepting them. The actual execution, running every script, reading every output, deciding what a confusing result meant, and choosing how to move forward, was done by me, using my own existing knowledge combined with what I learned along the way through this process.

This project was genuinely difficult for me, and I am disclosing that honestly rather than presenting it as if I already knew everything going in. What I can say with confidence is that I understand what was built and why, because I questioned, checked, and worked through each part myself rather than accepting it passively.

A.3 Output Validation

All AI-suggested code was run by me directly, and outputs were checked against expectations before being accepted. Several instances of this validation step caught real issues: a missing-value flag column that was documented as a decision but omitted when first building a database table (caught by directly querying PostgreSQL rather than trusting the script's success message), and a miscounted figure in an early decision log entry (5 vs. 7 flagged price outliers, caught by comparing the script's printed count against the written log).

A.4 Modifications Made

I made several changes during the actual analysis, not just to how things were worded. When the price outlier threshold was first just a placeholder number (€2,000), I pushed to look closer instead of accepting a round figure, this led to checking actual listing names and landing on €40,000 as the real, evidence-based cutoff. I also caught a missing-value flag that should have been in the database but wasn't, by checking pgAdmin myself, not because the AI noticed it first, and asked for it to be fixed. I chose rank-biserial correlation over Cohen's d as the main effect size once I understood why Cohen's d was getting distorted by the skewed price data.

A.5 Critical Assessment

I didn't accept results automatically. When over 27,000 rows got flagged as “duplicates” in the reviews data, I didn't assume that meant something was wrong, we checked the detailed reviews file together and confirmed they were real, separate reviews. When a regression showed beds having a negative effect on price, which didn't make sense at first, I asked for it to be looked into rather than just reported, and we traced it to overlap with bedrooms and accommodates. I also questioned relying only on Shapiro-Wilk to justify using non-parametric tests, since I remembered from my coursework that large samples can sometimes still work with standard tests through the Central Limit Theorem, this led to a better, corrected explanation in the final statistics write-up.